

Exercise

- Solve the dual problem to the Farmer's problem with selling of land and only wheat and rye

The problem itself:

Profit of 5786 for 91.4 hectares of wheat and 14.3 hectares of rye and 4.3 hectares sold

```
A=[1 1.5 -3 ; 1 2 0; 1 1 1];  
b=[100 120 110];  
-f=[-50 -85 0];  
lb=[0 0 0];  
ub=[110 110 110];  
[solution,profit,exitflag]=linprog(-f',A,b',[],[],lb',ub',opts);
```

The dual problem:

Fval=5786 for ys of 4.3, 32.9 and 12.9

```
[solution,profit,exitflag]=linprog(b',-A',-f',[],[],lb',ub',opts);
```

Meaning of dual

What are the y s?

They represent shadow prices for the cost of procuring the wheat, rye and land. It is hard to understand if you are not into economy.

We will understand the dual in the crystal plasticity problem – it is on our own turf....

Exercise

By means of MatLab:

Maximize $x_1 + x_2$ subject to the constraints $x_j \geq 0$ for all j and

$$x_1 + 2x_2 \leq 4$$

$$4x_1 + 2x_2 \leq 12$$

$$-x_1 + x_2 \leq 1$$

Formulate and solve the dual problem

Solution

```
A=[1 2; 4 2; -1 1];
```

```
b=[4 12 1];
```

```
lb=[0 0 ];
```

```
ub=[100 100 ];
```

```
-f=[-1 -1];
```

Max value is 3.3 for x1, x2 being 2.7 0.7

```
%dual
```

```
lb=[0 0 0 ];
```

```
ub=[100 100 100 ];
```

```
[solution,fval]=linprog(b',-A',-f',[],[],lb',ub',opts);
```

Min value of dual is 3.3 for y1, y2 and y3 being 0.33 0.17 0

Note: we have 3 ys and only 2 xs, i.e. need to add to lb and ub. If not MatLab adds $\pm\infty$ and you will get a different solution